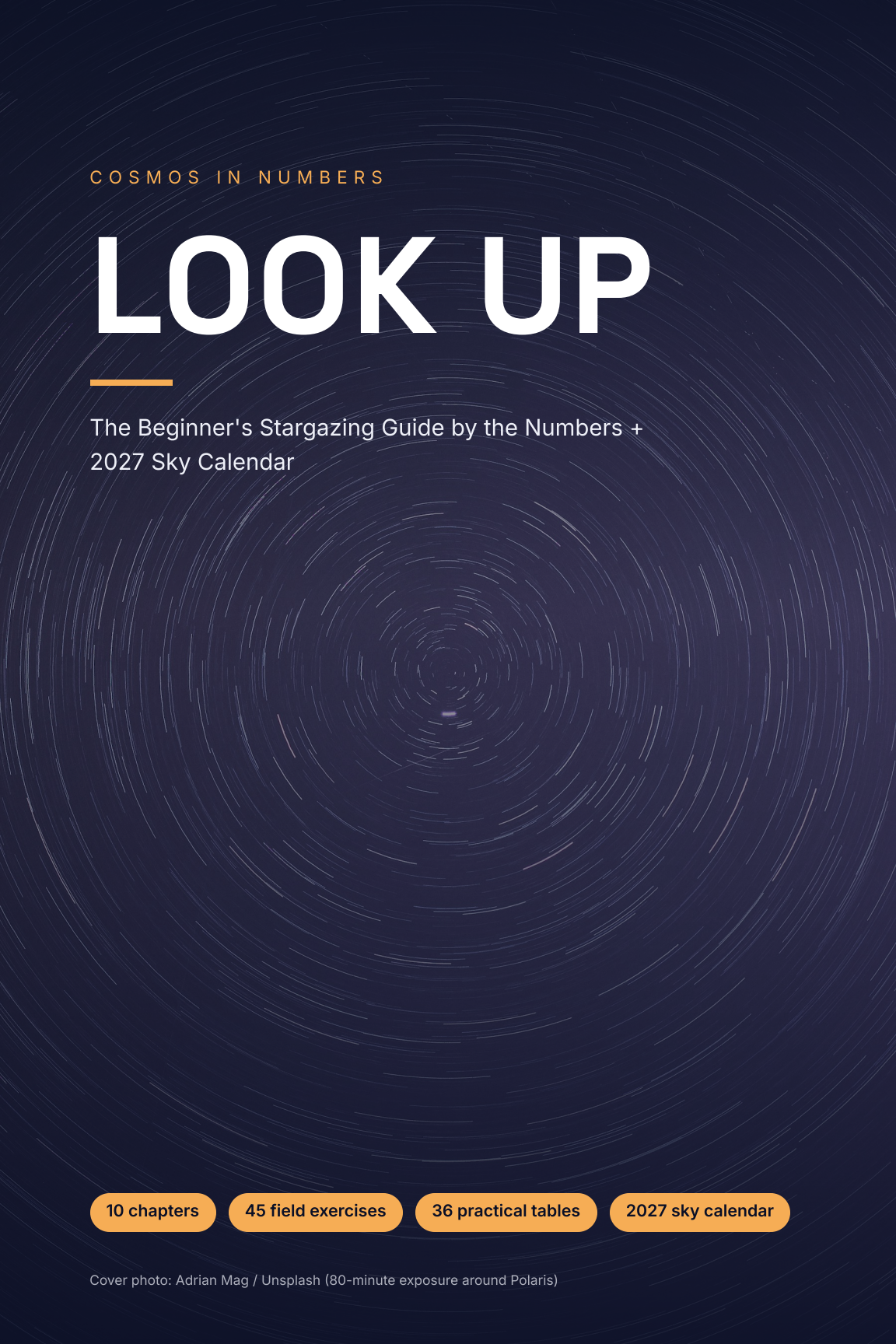
COSMOS IN NUMBERS

LOOK UP

The Beginner's Stargazing Guide by the Numbers + 2027 Sky Calendar



10 chapters · 45 field exercises · 36 practical tables · 2027 sky calendar

Cover photo: Adrian Mag / Unsplash (80-minute exposure around Polaris)

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CHAPTER 1

Your First Night Out

Twenty minutes, a dark corner and ten targets you can see tonight with your eyes alone.

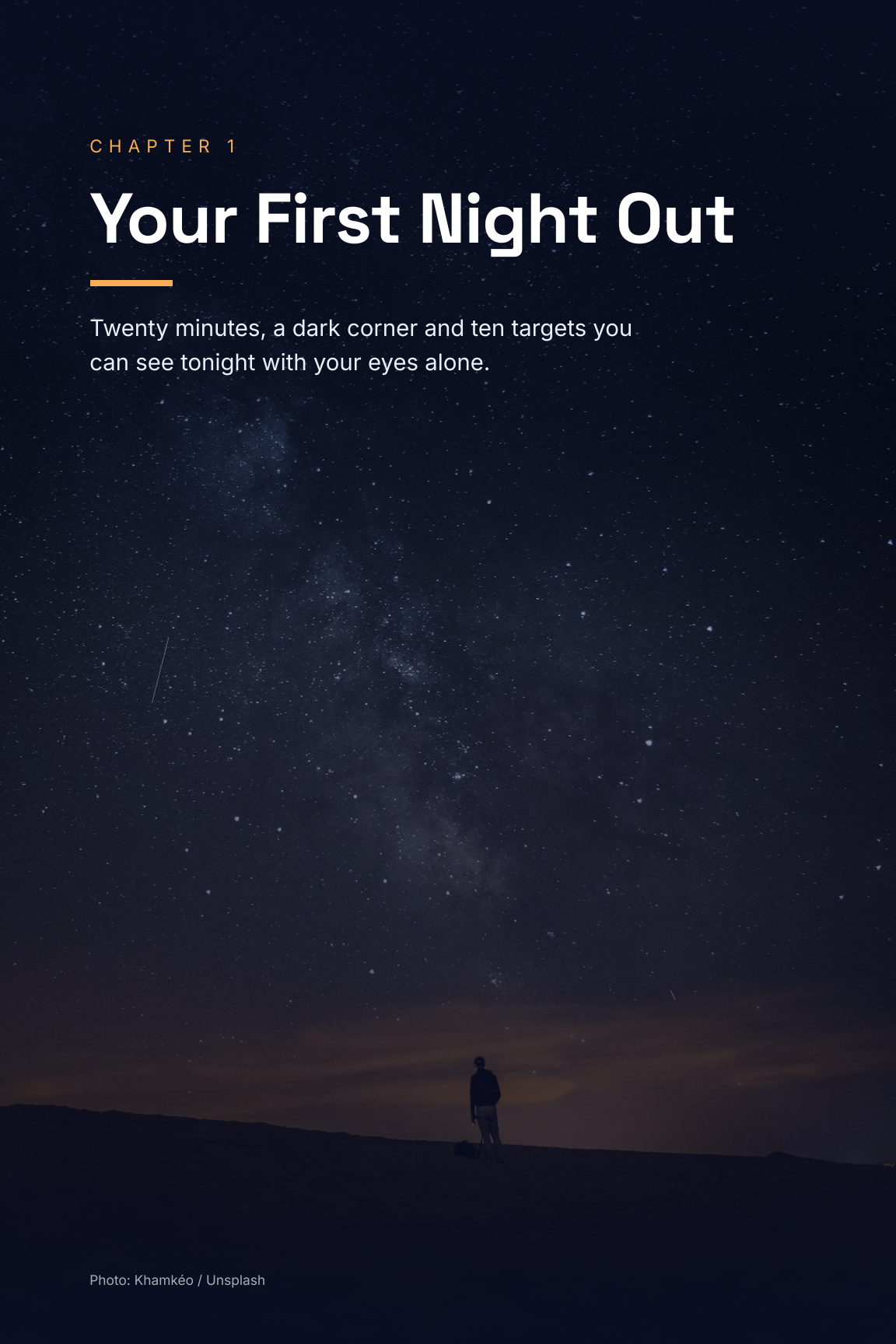


Photo: Khamkéo / Unsplash

Tonight, without buying anything, you can see light that left a galaxy two and a half million years ago. You just need twenty minutes, a dark corner and to know where to look.

Most people who say "I can't see anything up there" are not wrong about what they saw. They are wrong about how long they looked. They stepped out of a bright kitchen, glanced up for thirty seconds, counted a handful of stars and went back inside. Their eyes never had a chance to switch on. This chapter fixes that. By the end of it you will know how your eyes adapt to the dark, how many stars you can realistically expect to see from where you live, how to measure the sky with nothing but your hand, and which ten objects to look for on your very first night. No telescope, no app, no money.

We are going to treat your first night out the way a scientist would treat a first observation: with a plan, a few numbers and a notebook. That is the whole philosophy of this book. The sky is not vague or mystical. It is measurable, and once you start measuring it, it becomes much easier to see.

Your eyes are the first instrument

The human eye is a remarkable light detector, but it has two modes. In daylight and under bright artificial light, you rely mostly on your cone cells, which see colour and fine detail but need plenty of light. In the dark, a second set of cells, the rods, gradually takes over. Rods cannot see colour well, but they are far more sensitive to faint light. The switch from one system to the other is not instant. It is a slow chemical process: the light-sensit-

ive pigment in your rods, called rhodopsin, is bleached by bright light and needs time in darkness to rebuild.



A red light keeps your night vision intact while you move around. Photo: Marek Piwnicki / Unsplash.

That rebuilding is what astronomers call **dark adaptation**. Your pupils widen within seconds, but the real gain comes from the chemistry in the retina, and that takes time. Most of the improvement happens in the first 20 to 30 minutes, and sensitivity can keep creeping up for a while after that. In practical terms: if you go outside and judge the sky after one minute, you are judging it with a fraction of the sensitivity you will have half an hour later.

> **The 20-minute rule.** Give your eyes at least 20 minutes of real darkness before you decide what the sky looks like. Use that time to set up a chair, find north and let the faint stars appear one by one. They will.

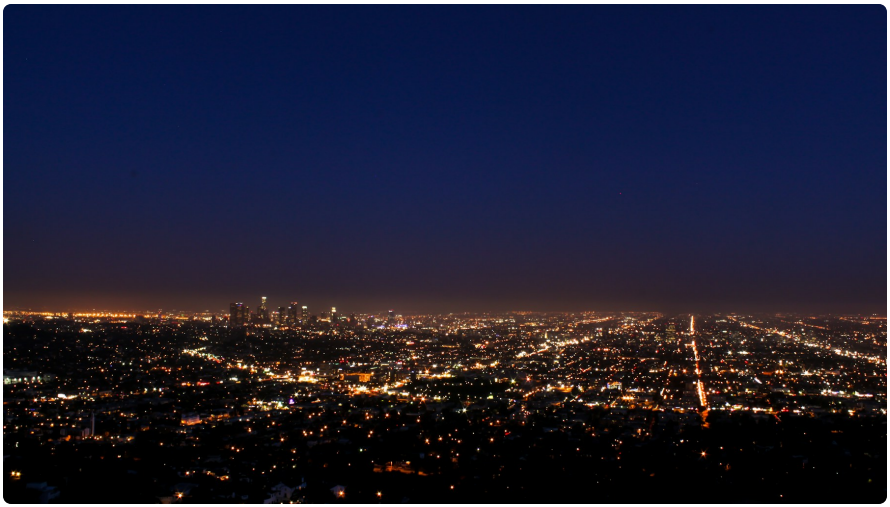
There is a catch. Dark adaptation is fragile. A single glance at a bright phone screen, a car's headlights or a white torch can undo much of it in seconds, and you are back at the start of the clock.

ively insensitive to deep red light, so a dim red torch lets you read a chart or write in a notebook without destroying your night vision. You do not need special equipment for this: a cheap headlamp with a red mode works, and so does a phone set to its night or red filter with the brightness turned to the minimum. If all you have is a normal torch, cover it with two layers of red cellophane or a red bike light and keep it pointed at the ground.

One more trick that sounds strange but works: **averted vision**. The centre of your retina, the part you use when you look straight at something, is packed with cones and has very few rods. The rods are more numerous slightly off-centre. So a faint star or a dim smudge of a galaxy is often easier to see if you look a little to one side of it rather than directly at it. Try it tonight on the faintest star you can find: stare at it, then shift your gaze a little to the side, and watch it brighten in the corner of your eye. Experienced observers do this without thinking. You will too, after a few nights.

How many stars can you really see?

Here is a number that surprises most people. Under a truly dark sky, far from any town, the whole sky holds roughly 9,000 stars bright enough for the human eye to see. That figure comes from catalogues of naked-eye stars such as the Yale Bright Star Catalogue, which lists around 9,100 stars down to about magnitude 6.5, the traditional limit of unaided vision. But you never see all of them at once. Half the sky is below your horizon at any moment, and stars close to the horizon are dimmed by the thick layer of air you look through. Under excellent conditions you might see somewhere around 2,000 to 3,000 stars in a single view.



Los Angeles by night: the stars are still there, drowned out by scattered artificial light. Photo: Benni Talent / Unsplash.

Now compare that with what most people see. From the middle of a large city you may count only a few dozen stars on a clear night. From a suburb, a few hundred. The stars have not gone anywhere. They are simply drowned out by the glow of artificial light scattered back down by the atmosphere. Astronomers have a scale for this, invented by the American amateur astronomer John Bortle and published in Sky & Telescope magazine in 2001. It runs from 1 to 9.

The Bortle scale, simplified.

Class 1–2, excellent dark sky: the Milky Way casts structure and detail, the sky looks crowded with stars, and you can barely see your own hand. Found in remote deserts, mountains, oceans and designated dark-sky parks.

Class 3–4, rural to rural/suburban transition: the Milky Way is obvious, with some glow from towns on the horizon. Many countryside locations.

Class 5–6, suburban: the Milky Way is faint or only visible overhead; the sky is grey rather than black near the horizon.

Class 7–9, bright suburban to inner city: the Milky Way is invisible, only the brightest stars and the planets show, and the sky glows orange or grey.

You do not need to drive to a Class 1 site to enjoy the sky. Every object in this chapter can be seen from a Class 6 or even Class 7 location. What the Bortle scale gives you is a way to set expectations and to measure progress. Later in the book (Chapter 8) we will look at how to find darker skies near you and how much difference a short drive can make. For now, a simple self-test will tell you roughly where you stand.

The Little Dipper test: measure your own sky

If you live in the Northern Hemisphere, the Little Dipper (part of the constellation Ursa Minor) is a perfect light-pollution meter. Its seven stars have a convenient spread of brightness. Polaris, the North Star, at the end of the handle, and the two stars at the far end of the bowl are fairly easy to see even from a suburb. The other four are fainter, and the faintest of them sits close to magnitude 5. How many of the seven you can see, after 20 minutes of dark adaptation, gives you a rough idea of your sky:



The Milky Way over a dark coast, with the glow of distant towns on the horizon. Photo: Evgeni Tcherkasski / Unsplash.

- **Only Polaris and one or two others:** bright suburban or city sky, about Bortle 7 or brighter.
- **Four or five stars:** typical suburban sky, around Bortle 5–6.
- **All seven, easily:** a genuinely dark rural sky, Bortle 4 or better.

From the Southern Hemisphere, where the Little Dipper is never visible, you can do the same thing with the stars of the Southern Cross (Crux) and its surroundings, or simply count the stars inside the four main stars of the Cross: the more faint stars you see inside and around it, the darker your sky. Whatever you use, write the result down with the date and place. It is your first real observation, and it gives you a baseline to compare against when you try a darker site.

Field exercise 1.1

Your sky score

Time: 25 minutes (20 of them dark adaptation) · Kit: notebook, red light

1. Go outside, switch off all lights and wait 20 minutes without looking at a screen.
2. Do the Little Dipper test (north) or count the faint stars inside the Southern Cross (south).
3. Write the number of stars, the date, the time and the place. Estimate your Bortle class from the box above.

Keep it: this is your baseline. Repeat the test at a darker site (Chapter 8) and compare.

Magnitude: the number that tells you how bright

Astronomers measure brightness with a scale called **magnitude**, and it runs backwards, which confuses everyone at first. The lower the number, the brighter the object. It goes back more than two thousand years to the Greek astronomer Hipparchus, who sorted stars into six classes: the brightest were "first magnitude", the faintest he could see were "sixth magnitude". In the nineteenth century the scale was made precise: a difference of 5 magnitudes is defined as exactly a factor of 100 in brightness, so each step of one magnitude is a factor of about 2.5.

Because the scale was extended in both directions, very bright objects have negative magnitudes. A few useful landmarks:

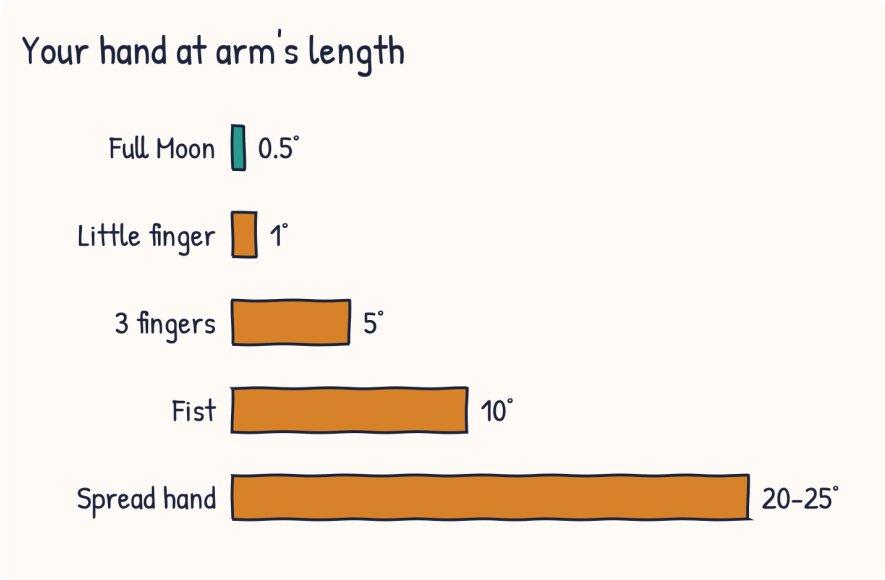
- **The full Moon:** about magnitude −12.7.
- **Venus at its brightest:** about −4.6, bright enough to cast faint shadows under a very dark sky.

- **Jupiter at its brightest:** close to −2.9.
- **Sirius, the brightest star of the night sky:** about −1.46.
- **A typical bright star such as Polaris:** about +2.
- **The faintest stars under a dark sky:** about +6 to +6.5.

You do not need to memorise these. What matters is the idea that brightness is a number, and that a difference of a few magnitudes is enormous. Sirius is roughly 25 times brighter than Polaris. Venus at its peak is around 15 times brighter than Sirius. When this book says that an object is "magnitude 4", you will know it is easy from a suburb, while "magnitude 6" means you need a dark site and patience.

Measure the sky with your hand

Star charts and astronomy apps describe positions and distances in the sky in degrees. The whole circle around you is 360 degrees, and from the horizon straight up to the point overhead, called the zenith, is 90 degrees. The good news is that you carry a surprisingly reliable measuring tool everywhere you go: your hand, held at arm's length. Because people with longer arms tend to have bigger hands, the angles work out about the same for most adults.



Hand-drawn diagram, Cosmos in Numbers. Angles are approximate.

> **Your hand at arm's length.**
>
> Little finger width: about **1°**. The full Moon is about half a degree across, so your little finger easily covers it.
>
> Three middle fingers together: about **5°**.
>
> A closed fist: about **10°**.
>
> Spread hand, tip of thumb to tip of little finger: about **20–25°**.

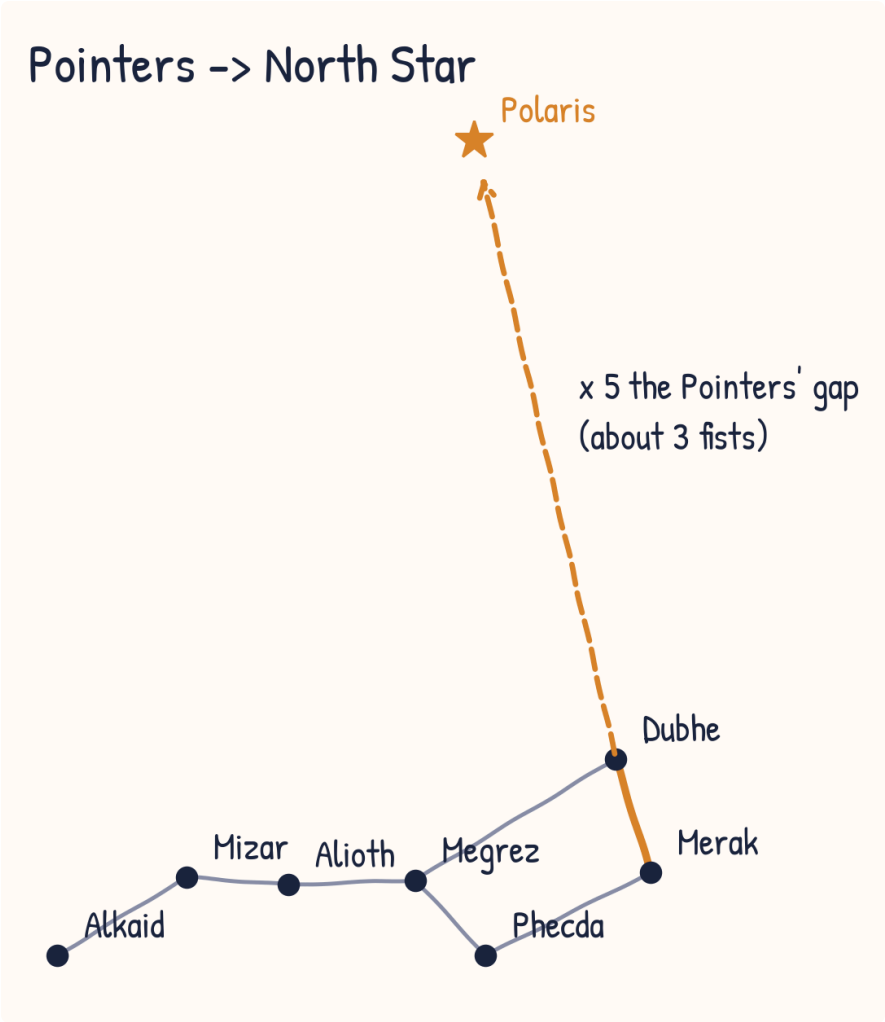
Try a quick calibration. The two stars at the end of the bowl of the Big Dipper, known as the Pointers (Dubhe and Merak), are about 5 degrees apart: roughly three fingers. The full length of the Big Dipper, from the end of the handle to the end of the bowl, is about 25 degrees: one spread hand. Once you trust your hand, every instruction like "look 10 degrees above the western horizon" becomes simple: one fist up from the horizon, towards the west.

Your hand also helps with one of the most common beginner surprises: the full Moon looks enormous when it rises, yet your little

finger still covers it. The Moon's size in the sky barely changes between rising and setting. The "Moon illusion" is a trick of perception, not of optics. Test it yourself next full Moon. It is a lovely first lesson in trusting measurement over impression.

Finding your way: north, south and the meridian

Before you can find anything, you need to know which way you are facing. In the Northern Hemisphere, the most reliable guide is Polaris. It is not the brightest star in the sky (that is Sirius), and it is only of medium brightness, but it has one special property: it sits less than one degree from the north celestial pole, the point around which the whole northern sky appears to turn. While every other star rises and sets, Polaris stays almost fixed above the north point of your horizon. Its height above the horizon, measured in degrees, is approximately equal to your latitude. From New York, at about 41 degrees north, Polaris sits about four fists above the northern horizon. From Rome, at about 42 degrees, almost the same.



Hand-drawn diagram from J2000 star positions, Cosmos in Numbers.

To find it, use the Big Dipper. Follow the line from Merak to Dubhe, the two Pointers at the end of the bowl, and extend it about five times their separation, roughly 28 degrees or a little less than three fists. You will land on Polaris. Now you know where north is, and therefore south, east and west too.

In the Southern Hemisphere there is no bright star at the pole. Observers there use the Southern Cross and the two bright Pointers next to it, Alpha and Beta Centauri. Extend the long axis of the Cross about four and a half times its length towards the south and you are close to the south celestial pole; draw a line at right angles from the middle of the two Pointers and the two lines meet near the same point. Drop straight down from that spot to the horizon, and you are facing south.

There is one more concept worth knowing on night one: the **meridian**, an imaginary line that runs from the north point of your horizon, through the zenith, to the south point. Every star and planet reaches its highest point in the sky when it crosses the meridian. Objects are easiest to see when they are high, because you are looking through less air. So whenever you can, observe things near the meridian rather than low on the horizon.

The first ten targets: no equipment required

Here is a list of ten things a beginner can find with the naked eye. You will not see all ten every night, because the sky changes with the seasons and the planets move, but over a few months of casual observing you can tick off every one. Each has a fact attached, because knowing what you are looking at changes how it feels to see it.



The full Moon, about 384,400 km away: its light reaches you in about 1.3 seconds. Photo: Mike Petrucci / Unsplash.

1. The Moon. The easiest target in the sky and the most underrated. It is about 384,400 kilometres away on average, so its light reaches you in about 1.3 seconds. Look at it a few days after new Moon, when it is a thin crescent in the evening sky, and you may notice the dark part of the disc faintly glowing. That glow is *earthshine*: sunlight reflected off the Earth onto the Moon and back to your eyes. You are seeing your own planet's light, bounced off another world.

2. Venus. When it is visible, it is unmistakable: a brilliant, steady white light low in the west after sunset or in the east before sunrise. It is the brightest natural object in the night sky after the Moon. Its brightness comes from its thick cloud cover, which reflects most of the sunlight that hits it. It never strays far from the Sun in our sky, so you will never see it high overhead at midnight.

3. Jupiter. The second-brightest planet most of the time, a cream-coloured point that does not twinkle as much as the stars around it. It is the largest planet in the Solar System, with a diameter

about 11 times Earth's. With binoculars held steady, you can often see up to four tiny points of light lined up next to it: the four large moons discovered by Galileo in 1610.

4. Mars. Its colour gives it away: a distinctly orange-red point. Its brightness changes dramatically from year to year, because its distance from Earth varies enormously. At its closest approaches it can rival Jupiter; at its farthest it is no brighter than an ordinary star.

5. Saturn. A steady, yellowish point, less dazzling than Jupiter but still among the brighter "stars" when it is up. To the naked eye it is just a point. But even a small telescope reveals its rings, and that first view is one of the great moments of amateur astronomy. Chapter 4 explains when and how.

6. Orion. The most recognisable constellation in the world, visible in the evening from both hemispheres from about November to March. Look for three bright stars in a short straight line, Orion's Belt, framed by four brighter stars in a rough rectangle. The top-left star (as seen from the Northern Hemisphere) is the red supergiant Betelgeuse; the bottom-right is the blue-white Rigel. The difference in colour between them is easy to see once you know to look for it, and it tells you their surface temperatures are very different.

7. The Orion Nebula. Below the Belt hangs a short line of fainter stars, Orion's Sword. The middle "star" of the Sword is not a star at all. Under a reasonably dark sky it looks slightly fuzzy, and it is: you are looking at a vast cloud of gas and dust where new stars are being born, about 1,300 to 1,350 light-years away. It is one of the few nebulae you can see with the naked eye.

8. The Pleiades. A tight little cluster of stars, often mistaken for a tiny dipper, visible in the evening sky from late autumn to early

spring. Most people with average eyesight count six or seven stars; with binoculars you will see dozens. They are a family of young stars that formed together, about 440 light-years from us. Their Japanese name is Subaru, which is why the car company's logo shows a cluster of stars.

9. The International Space Station. If you see a bright, steady point of light moving smoothly across the sky over a few minutes, without blinking, it is very likely the International Space Station. It orbits about 400 kilometres up, circles the Earth roughly every 90 minutes and shines by reflecting sunlight, which is why it is visible shortly after sunset or before sunrise. NASA's free Spot the Station service tells you exactly when it will pass over your location. Watching it for the first time, knowing there are people aboard, is a strangely moving experience.

10. The Andromeda Galaxy. The most distant thing you can see with your eyes alone. Under a dark sky, in the autumn and winter evenings of the Northern Hemisphere, it appears as a faint, elongated smudge in the constellation Andromeda. NASA puts its distance at about 2.5 million light-years. The light reaching your eye tonight left it long before our own species existed. Finding it takes practice and a dark site, so treat it as the graduation target of this first chapter.

Three star patterns for every season

Constellations are the street names of the sky. There are 88 official ones, defined by the International Astronomical Union in 1922 and given precise boundaries in 1930, so that every point of the sky belongs to exactly one constellation. You do not need to learn all 88. Experienced observers navigate with a handful of bright, easy patterns and then "star-hop" from them to fainter things.

Here is a starter set of three patterns per season for the evening sky of the Northern Hemisphere, with notes for southern observers. Learn one season at a time, and within a year you will know the whole sky.

Winter evenings (December to February). Start with **Orion**, which you already met. Follow the line of the Belt down and to the left (as seen from the north) and you reach **Sirius**, in Canis Major, the brightest star of the night sky, about 8.6 light-years away. Follow the Belt up and to the right and you reach the orange star Aldebaran in **Taurus**, and then the Pleiades. Those three steps, Orion to Sirius and Orion to Taurus, give you the backbone of the winter sky. From the Southern Hemisphere, the same patterns are visible on summer evenings, upside down.

Spring evenings (March to May). The **Big Dipper** rides high in the north. Use its handle as a curved arrow: "arc to Arcturus", the bright orange star in Boötes, then "spike to Spica", the blue-white star in Virgo. Between them, look for the backwards question mark that forms the head of **Leo**, the lion, with the bright star Regulus at its base. Southern observers see Leo and Virgo well placed in their autumn evenings, while the Big Dipper stays low or below the horizon.

Summer evenings (June to August). Look for the **Summer Triangle**, three bright stars from three different constellations: Vega in Lyra, Deneb in Cygnus and Altair in Aquila. It is huge, spanning several fists, and almost overhead late in the evening. If your sky is dark enough, the Milky Way runs right through it. Low in the south lies **Scorpius**, with the red supergiant Antares at its heart, and next to it the teapot shape of **Sagittarius**, which points towards the centre of our galaxy, about 26,000 light-years away.

From the Southern Hemisphere, Scorpius and Sagittarius pass high overhead in the winter months and are spectacular.

Autumn evenings (September to November). Find the **Great Square of Pegasus**, four moderately bright stars forming a large, nearly empty square. From its top-left corner (as seen from the north) a curved chain of stars marks **Andromeda**; following it is the classic way to find the Andromeda Galaxy. Higher up, the W-shaped **Cassiopeia** sits on the opposite side of Polaris from the Big Dipper, so when one is low the other is high. Southern observers see Pegasus low in the north and get the two Magellanic Clouds, satellite galaxies of the Milky Way, high in the south: naked-eye galaxies that northern observers never see.

> **Star-hopping in one sentence.** Start from something you know, move a measured number of fists in a known direction, and look for the next landmark. Every observer in history found their way around the sky exactly like this, and so will you.

Do not worry if the patterns look different from the drawings in books. Constellation figures are human inventions, and many of them look nothing like the animals or heroes they are named after. What matters is that you can recognise the shapes the stars actually make. A good tip is to sketch what you see in your notebook, even badly. Drawing forces you to notice the angles, the spacing and the brightness of the stars, and once you have drawn a pattern, you rarely forget it.

Field exercise 1.2

Ten-target checklist

Time: over several nights · Kit: this list, a pencil

- ±0 The Moon (and earthshine on a crescent)
- ±0 Venus
- ±0 Jupiter
- ±0 Mars
- ±0 Saturn
- ±0 Orion
- ±0 The Orion Nebula
- ±0 The Pleiades
- ±0 The International Space Station
- ±0 The Andromeda Galaxy

Goal: five in your first week, all ten within a season.

Your first night: a step-by-step plan

Plans make nights succeed. Here is a simple routine you can follow tonight or the next clear evening. It takes about an hour, and you can do it from a garden, a balcony or a park.

Step 1: Check the sky and the Moon (5 minutes, indoors). Look at a weather forecast for cloud cover. Then check the Moon's phase. A bright Moon is great for observing the Moon itself but washes out faint objects. For a first night, any phase is fine; for Andromeda or the Milky Way, aim for the week around new Moon.

Step 2: Choose your spot. Find the darkest place you can reach easily, with a wide view of the sky and no lights shining directly at you. Even moving from a lit front garden to the shadow of the house makes a difference. If you can, choose a spot with a clear view to the south (Northern Hemisphere) or the north (Southern Hemisphere), where most of the action happens.

Step 3: Prepare your kit. Warm clothes (it is always colder than you think after an hour of standing still), a reclining chair or a blanket, a red light, a notebook and a pencil. A paper star chart or a printed sky map for the current month is ideal. If you use a phone app, switch on its red or night mode before you go outside.

Step 4: Start the 20-minute clock. Turn off your lights, sit down and look around. Do not judge the sky yet. Use this time to find the brightest objects: the Moon, any very bright "stars" that might be planets, and a few recognisable patterns.

Step 5: Orient yourself. Find north or south using the methods above. Note which constellations you can see and roughly where they are.

Step 6: Run the light-pollution test. Count the stars of the Little Dipper or the Southern Cross. Write the number down.

Step 7: Hunt the list. Go through the ten targets and tick off what is visible tonight. For each one, write down the time, where it was in the sky (for example, "two fists above the eastern horizon") and anything you noticed.

Step 8: Wait for the ISS or a meteor. If you checked Spot the Station beforehand, be ready at the right time. Otherwise, spend the last ten minutes lying back and watching a wide patch of sky. On an average night, outside the big meteor showers, you will typi-

cally catch a few random "shooting stars" per hour from a dark site.

> **Write it down.** Observers who keep notes improve much faster than those who don't. The printable observing log included with this book has a page for every night: date, place, Bortle estimate, sky conditions, what you saw. After 30 nights you will have a record of your own progress, and you will be surprised how much you have learned.

Seven mistakes that ruin first nights

Almost every beginner makes some of these. Knowing them in advance saves you a lot of frustration.

Looking at your phone. The number one killer of dark adaptation. Every time you check a bright screen, you reset your eyes. Red mode, minimum brightness, or keep it in your pocket.

Giving up too soon. Five minutes is not enough. The sky you see after 30 minutes is a different sky.

Going out under a full Moon to look for faint things. The full Moon brightens the whole sky. It is the best night to study the Moon and the worst night for galaxies.

Starting with a telescope. Many beginners buy a telescope first and are disappointed. Without knowing the sky, they cannot find anything, and a narrow field of view makes it harder. Learn the sky with your eyes, then add binoculars. Chapter 7 explains why binoculars are the smartest first purchase.

Expecting Hubble images. Photographs of nebulae and galaxies are long exposures, often taken with large telescopes and processed in colour. Your eye sees faint grey shapes. That is not a failu-

re: it is the real light from the real object, arriving at your eye right now.

Dressing for the temperature on the forecast. Standing still at night, you lose heat fast. Dress as if it were 10 degrees Celsius colder than the forecast, and you will stay out longer.

Trying to see everything at once. Pick three or four targets per night. Depth beats breadth. The best observers spend a long time looking at a few things.

Safety and courtesy under the stars

Astronomy is a very safe hobby, but a few rules matter. Never look at the Sun, with or without binoculars or a telescope, unless you are using a proper certified solar filter or a dedicated solar viewer; serious eye damage can happen in seconds, and it is painless while it happens. When observing away from home, tell someone where you are going, carry a charged phone, and choose places where you are allowed to be at night. Respect private land and protected areas. And if you share a dark site with other observers, keep white lights off and car headlights pointed away. Ruining someone's dark adaptation after their 30-minute wait is the one real sin of the hobby.

What you have achieved tonight

If you followed the plan, you have done more than "look at the stars". You have calibrated your eyes, measured your sky's darkness, learned to read brightness as a number and distance in the sky as degrees, found north or south, and identified real objects at distances ranging from 1.3 light-seconds (the Moon) to 2.5 million

light-years (Andromeda). That is a range of distances of more than ten trillion to one, crossed in a single evening with your own eyes.

But you probably also noticed something unsettling: by the end of the hour, the stars were not where they had been at the beginning. Orion had climbed higher, or the Big Dipper had swung around. The sky moves, all night and all year, and that movement follows simple, predictable rules. Once you understand them, you will be able to predict what will be up at any time on any night of the year. That is where we go next.

Sources for this chapter. NASA Science, Moon facts (average Earth–Moon distance 384,400 km) and Andromeda Galaxy (M31, about 2.5 million light-years); NASA, International Space Station facts and Spot the Station (spotthestation.nasa.gov); NASA Planetary Fact Sheet, nssdc.gsfc.nasa.gov/planetary/factsheet (Jupiter diameter); Yale Bright Star Catalogue, 5th revised edition (about 9,100 stars to magnitude 6.5); International Astronomical Union, The Constellations (88 constellations, boundaries set in 1930); NASA Science, Sirius (about 8.6 light-years) and the Milky Way's centre (about 26,000 light-years); J. E. Bortle, "Introducing the Bortle Dark-Sky Scale", Sky & Telescope, February 2001. Magnitudes of Venus, Jupiter, Sirius and the full Moon are standard published values; distances of the Pleiades and the Orion Nebula are rounded from published parallax-based measurements.

Keep Going

Cosmos in Numbers is an independent project that explains space with real data: real images from NASA and ESA, charts built from public scientific data, and every number linked to its source. No hype, no invented facts.

Keep going

- **Watch:** short, sourced space explainers and longer videos at **youtube.com/@CosmosinNumbers**.
- **Follow:** @cosmosinnumbers on TikTok and @cosmos.innumbers on Instagram for one real number about space every day.
- **Join:** the "Cosmos in Numbers – Space by the Data" group on Facebook: share your observing logs and questions.
- **Read:** the website cosmosinnumbers.github.io.

Your bonus material

- The printable observing log that follows this page (print as many copies as you like for personal use).
- The **Cosmos in Numbers Data Poster Pack**, if it is included in your bundle: 14 print-ready charts of the planets, exoplanets and the spacecraft leaving our Solar System.

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